

SOP-P201-MAINT-001 Rev.5 · P-201 Pump Maintenance Procedure (2024)

BPCL Mathura Refinery · Rotating Equipment Division

Document Control

Revision: 5. Date: 10-Jan-2024. Author: R. Gupta (Lead Mechanical Engineer). Approved by: A.K. Singh (Head, Reliability Engineering). Supersedes: Rev.2 dated March 2019. Status: ACTIVE · CURRENT APPROVED VERSION.

1. Scope

This procedure covers planned maintenance for P-201 Crude Oil Transfer Pump. Updated per API 610-12th Edition (2021) and BPCL Engineering Standard BES-ME-007 Rev.3. All previous revisions (Rev.1 through Rev.4) are superseded and must be destroyed.

2. Torque Specifications · REVISED

IMPORTANT: Torque values updated per API 610-12th Edition (2021) · OEM Flowserve issued Service Bulletin SB-PVXM-2023-04 revising coupling bolt torque. Coupling bolts (M24 grade 8.8): TORQUE TO 95 Nm (REVISED from 80 Nm in Rev.2). Bearing housing end cover bolts (M16): 45 Nm (unchanged). Baseplate hold-down bolts (M30): 180 Nm (unchanged). Casing flange bolts (M20): 95 Nm (unchanged).

3. LOTO Sequence · REVISED per OISD-105 Section 6.4

SAFETY CRITICAL: LOTO sequence revised per OISD-105 (2022 revision) Section 6.4 and incident investigation report IIR-BPCL-MTR-2022-019. LOTO STEP B: First close DISCHARGE valve V-201B completely (handwheel, 8 turns CW). LOTO STEP A: Then close SUCTION valve V-201A completely (handwheel, 5 turns CW). NOTE: Sequence B then A is now mandatory · reverse of previous Rev.2. Closing discharge first prevents reverse flow and cavitation during isolation. LOTO STEP C: De-energize MCC-2 breaker CB-201 and lock with personal padlock. LOTO STEP D: Attach danger tags at V-201B handwheel, V-201A handwheel, and MCC-2. LOTO STEP E: Open vent valve V-201C to release casing pressure. LOTO STEP F: Test for zero energy · verify 0 bar on pressure gauge PI-201.

4. PPE Requirements

Chemical resistant gloves (nitrile, minimum 8mil thickness · upgraded from 6mil). Full face shield (not safety glasses alone for crude oil service). Steel-toed safety boots with chemical resistant sole. Flame-resistant coveralls (Category 2 minimum, Cat 3 preferred). Hard hat. Hearing protection. H2S monitor mandatory · crude service may contain up to 50 ppm H2S.

5. Bearing Replacement · Updated

Use SKF TMHP 10 bearing puller or equivalent. Do not impact bearings. Clean housing with approved solvent (Exxon Varsol 60 or equivalent). Install bearing using SKF THAP 200 hydraulic fit method. Grease quantity increased: Apply Mobil SHC 100 grease (20g per housing) · increased from 15g per Flowserve Service Bulletin SB-PVXM-2023-04. Document bearing serial number in SAP PM notification.